

Reproduction of Zivich & Breskin's Table 3

1,000 simulated datasets (sim.id 1–1000), $n = 3000$; truth = -0.1081508 .
 Changes from the paper: g-computation CIs use the empirical SE (no bootstrap);
 single cross-fit (SC, 5 splits, 1 partition) replaces double cross-fit.

	Bias	RMSE	ASE	ESE	CLD	Coverage
G-computation						
True	+0.000	0.017	–	0.017	0.068	94.8%
Main-effects	-0.022	0.028	–	0.018	0.069	76.8%
Machine learning	+0.026	0.031	–	0.017	0.065	64.0%
IPW						
True	+0.007	0.025	0.025	0.024	0.097	94.4%
Main-effects	-0.022	0.032	0.023	0.023	0.091	86.7%
Machine learning	+0.011	0.024	0.023	0.021	0.089	93.7%
AIPW						
True	+0.000	0.021	0.020	0.021	0.078	94.1%
Main-effects	-0.016	0.026	0.020	0.020	0.077	85.4%
Machine learning	+0.004	0.020	0.017	0.019	0.065	90.8%
TMLE						
True	+0.000	0.021	0.020	0.021	0.077	93.8%
Main-effects	-0.017	0.025	0.019	0.018	0.076	85.9%
Machine learning	-0.001	0.020	0.016	0.020	0.065	90.1%
SC-AIPW						
True	+0.000	0.025	0.024	0.025	0.094	95.1%
Main-effects	-0.011	0.037	0.032	0.035	0.125	90.4%
Machine learning	-0.003	0.024	0.022	0.023	0.087	93.5%
SC-TMLE						
True	+0.003	0.025	0.023	0.024	0.090	94.6%
Main-effects	-0.017	0.026	0.028	0.020	0.109	92.1%
Machine learning	+0.001	0.020	0.020	0.020	0.080	93.8%

RMSE: root mean squared error; ASE: average estimated standard error; ESE: empirical standard error (SD of the 1,000 estimates); CLD: mean confidence-limit difference (interval width);

Coverage: fraction of 95% CIs containing the truth.